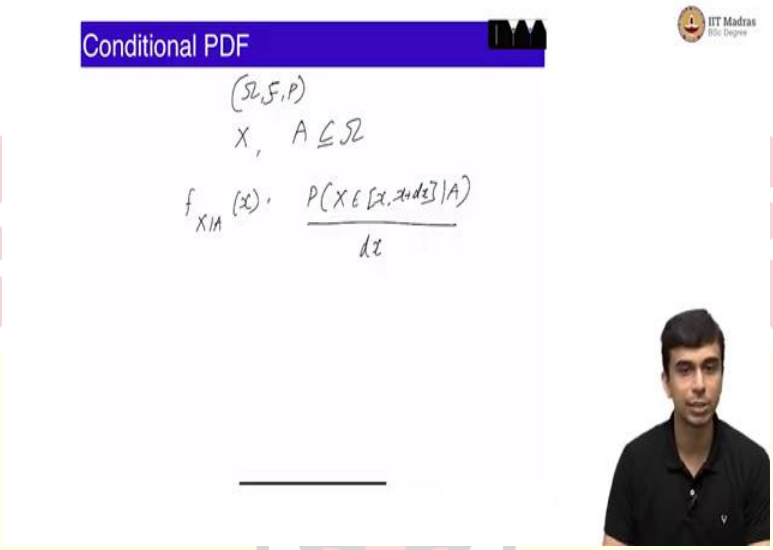


IIT Madras

ONLINE DEGREE

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Conditional pdf

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The same way that we had the conditional probability mass functions for discrete random variables. We can have a conditional probability density function for continuous random variables. What is that?

So, let us say you have the experiment given by the tuple omega f, p and you have a random variable x, and let us say you also have an event A, which is all events are simply subsets of omega and it also has to belong to f, but for now let us say you have this.

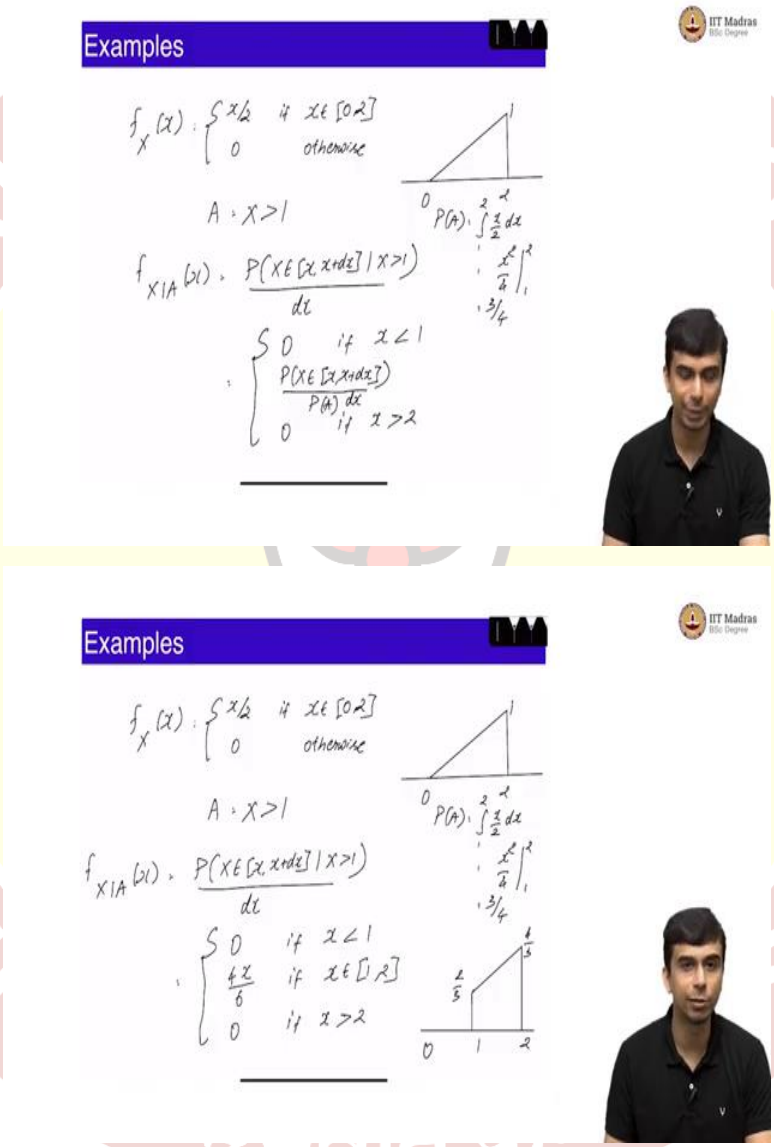
Conditional pdf of x is simply as follows, f x given A of x is simply the probability of x belonging to x to x + d x given A divided by d x, you see the conditional probability, a density function is the same as standard pdf, just that instead of probability of x belonging to x + d x, you have probability of x belong to x + d x given A.

And because A is an event and this is also an event this as a whole is a valid probability, this is a whole is a valid argument for a probability function. And have a d x here, the same way that you have for standard pdf.

How do you interpret this? That is A is this is f x given A of x is still actually a valid density function also, why is that? Well, you can see, that if you integrate are essentially, what the

conditional probability density function does is it captures the probability density function of x, given that A has happened. A is some information about the randomness of the experiment, but it is does not completely, let us say give you the information about x.

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Let us see a simple example here. Let us say you have x is distributed as the as follows x, f_X(x) is equal to x by 2, if x belongs to [0, 2] and 0 otherwise. So, we already seen that, this is a valid density function. And how will this look like? This will look as follows, value of 1 here.

So, this is the density, this is how the density function of x is going to look like. Let us have a very simple event here, the event A is simply x > 1, this is the event. And you that what does the

conditional probability density function now indicate, this is the question, what is the conditional probability density function of x?

So, even though now you happen to know that x > 1, you do not know completely x and we do not know x completely, because it can take anything between 1 and 2. But what is the density function of x. If you did not know x, this particular function effectively represents the uncertainty of x.

So, higher values mean that it is more likely to be there and lesser values mean that it is less likely to be there. But given that you have observed A, what is, how does the density function change, for 1 you should know that immediately, that the value between 0 to 1 has to go to 0.

Why is that? Because you have observed already that x > 1, because you have already observed that x is > 1, the density of the random variable x is in the range of 0 to 1 has to be, has to become 0 now.

But between 1 and 2, what is going to happen? Between 1 and 2, the it has to increase. So, let us say you had some 70 % chance of before observing x, let us say you have some 70 % chance of x to be being greater than 1, given that you have observed x to be > 1, there is a 100 % chance, that x > 1.

So, effectively that has to 70 % has to increase to 100 %, and the 30 % essentially has to go to 0 %. But how is that expressed in a concrete way, I am just giving some approximate numbers here, but we will do the exact I think, what is f x given A of x, first of all that would immediately give you, that by definition, it is going to be probability of x belonging to x to, x + d x given A by d x, in this case x is simply x > 1/2, x > 1.

This is f x given a of x, will get the easy things out first, it is going to be equal to 0, if x is less than 1, it is going to be equal to 0, if x > 2. The interesting value is between 1 and 2. Why is it going to be 0, which x > 2? Because x itself has nothing outside 0 to 2.

So, given A also it does not change much. So, the density of x given A is also equal to 0 of x > 2. This is a consequence of A. So, because you observed x is to be > 1, the density of x given a of x is going to be 0. But between 1 and 2 what is going to happen? So, between 1 and 2 this is you can use the law of conditioning can simply use the definition of conditional probability.

So, which is probability of x belonging to x + dx, given x > 1 is simply given by probability of x belonging to x + d x and x > 1 divided by probability of x > 1. But when small x is in the range of 1 to 2 probability of x belonging to x + d x, x to x + d x and probability and x > 1 is exactly the same as probability of x belonging to x to x + d x, why is that? Let us say for example x is 1.2.

What the joint probability would give you is the probability of x belonging to [1,2] to 1.2 + d x and x > 1. If x belongs to [1,2] and 1.2 + d x, it is clearly > 1. So, the and is basically useless, it is a vacuous statement.

So, now you have that would be the probability of x belonging to x to x + d x divided by probability of a by d into d x, of course you have a d x also. Let us just rewrite that, what is probability of a probability of A, probability of A would simply be the probability that x > 1, what is that? Well, that is exactly equal to the area under this curve.

So, probability of A simply equal to integral 1 to 2, x by 2 d x. So, that is simply x squared by 4 in the range of 2 to 1, which is equal to 1 - 1 by 4, which is 3 by 4. What this says is, a priori the probability of A happening is 3 by 4, 3 by with 75 % chance x > 1.

But there is a difference between a priori and posteriori, which is a priori means, that you have not observed the event A, posteriori means, that you have observed the event A. So, A has happened. So, you are effectively going to multiply probability of x belong to x + d x by 4 by 3.

So, that will essentially be x by 2 into 4 by, x by 2 divided by 3 by 4, which will be 4 x by 6. So, what will effectively happen is that, within this range, the density function is going to increase, but within this range 0 to 1, it is going to become 0.

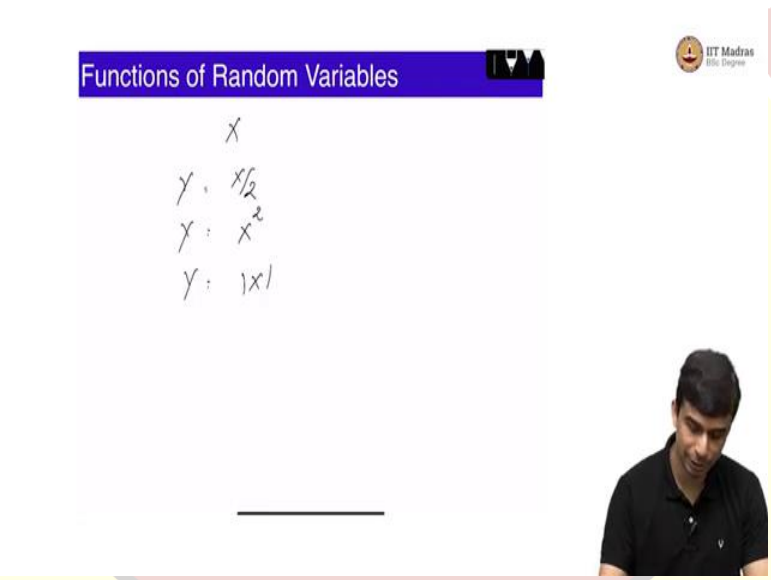
So, if you drew a plot of f x given A, it would look like this, draw from scratch, it is going to be 0, 1, 2, let us say, at 1 it is going to be 4 by 6, at 2 it is going to be 8 by 6. So, the value here it is 4 by 6, or 2 by 3, this is 2 by 3 and this is 8 by 6, or 4 by 3.

So, you can see that, this value is exactly not exactly the same as this. So, here it was half and 1, half and 1 has become 2 by 3 and 4 by 3. But that is simply because of the multiplication by 4 by a division by 3 by 4. So, this trend remains the same, the trend between 1 and 2 is still linear for the f x given A. But the values are different, because you are effectively divided by probability of A.

And here is also an example of the case. Now, you can see that the density function can be > 1, it can take values > 1, as long as the area of, as it integrates to 1, the integration to 1 is more is the important property.

So, this is a conditional probability density function and this is pretty powerful tool and given and you can have a random variable and you can ask the question given this event A has happened, what is the conditional, what is the density function of this random variable x, that is given by the conditional density function.

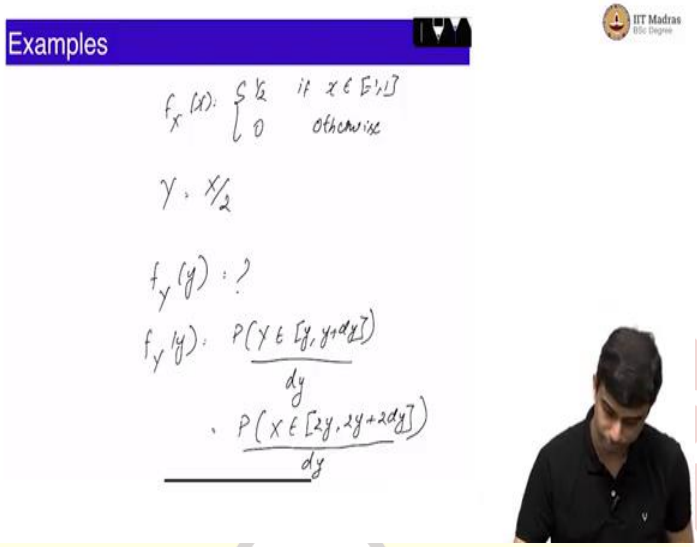
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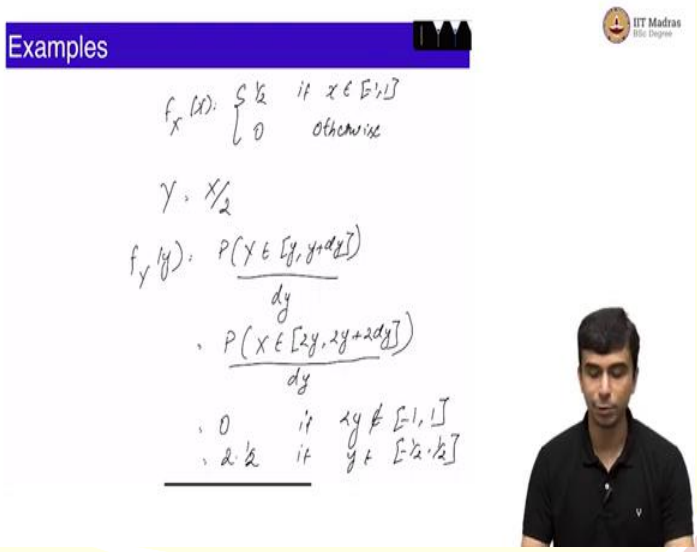


Now, we will move on to another topic, which is functions of random variables. Let us say you have the same thing, let us say you have random variable x, in the prob experiment given by omega f p, you can have a function of this random variable. Let us say take the simplest kind of function, let us say y is equal to x by 2, or let us say y is equal to x square or y is equal to absolute value of x.

So, these are all functions of the random variable x, these themselves would be valid random variables. So, y would also be a random variable, because it is a function of random variable. And how do you get the density function of this new random variable. Let us take the simplest example here.

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And let us actually do all the examples. Let us take the case where x is uniform in the range of let us say - 1 to 1, what does that mean, f x of x is equal to half, if x belongs to - 1 to 1 and 0 otherwise, clearly you can see this integrates to 1.

Now, y in this case let us say x by 2, can ask the question, what is the density function of this new random variable y. What is f y of y? That is the question and you can just use simple logic here, what is f y of y is probability of y belonging to y + sorry, y 2 y + d y divided by d y.

But y is exactly equal to x by 2, which means that this is exactly equal to probability of x belonging to x by 2 belonging to y + d y or x belonging to 2 y to 2 y + 2 d y divided by d y.

So, you have let us create some space, what is the probability that x belongs to 2 y to 2 y + 2 d y. So, here is where the range of the thing matters where x the probability of x belonging to x 2 x + d x is going to be nonzero, only if x is in the range of - 1 to 1. So, when 2 y is in the range of - 1 to 1 this is going to be nonzero.

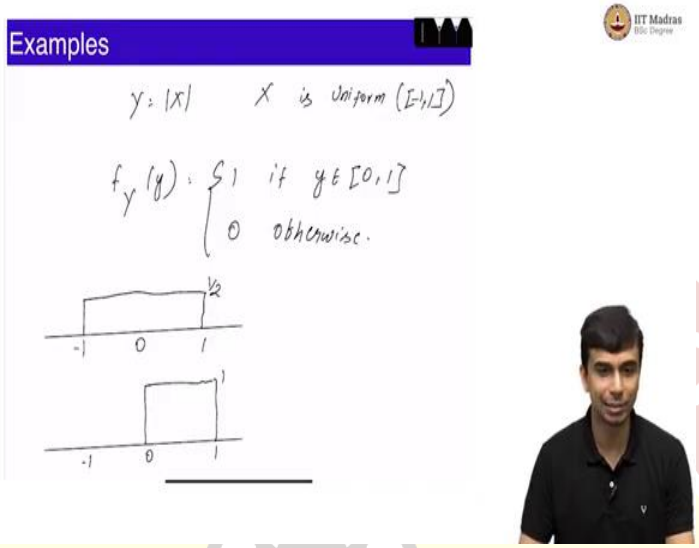
So, this is going to be equal to now note that here it is 2 d y. So, you have to be careful here, this is going to be 0 first of all, if 2 y does not belong to - 1 to 1, or in other words it is going to be 0, if y does not belong to - 2 to 2. And sorry, y does not belong to - half to half and that makes sense.

So, x is uniform - 1 to 1, you should immediately guess that y is going to, y is equal to x by 2, y is uniform - half to half. But we are going to do this just in a painful way. But outside - half to half, it is going to be 0.

But if in the range of - half to half, what is the value? The value is going to be equal to 2 sorry, 2 times half, if y is in the range of - half to half -, y is, why are you getting a 2 here, this half is simply because of the consequence of f x of x, this 2 is a consequence of 2 d y.

So, this is probability of x belonging to y 2 y + 2 d y. So, that is divided by d y that is going to be equal to 2 times half, that is the reason why you have 2 into half, which is equal to 1. So, which we have the density of y is equal to 1, if in the range of - half to half and 0 otherwise. So, very simple example

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Let us take a slightly more complicated example. Let us say y is equal to absolute value of x, let us take the same thing where x is uniform - 1 to 1. What is the density function of y going to look like? You can, you can use simple logic again, that is probability of y sorry, f y of y is going to be equal to 1, if y belongs to 0 to 1 and 0 otherwise. Why is that? Well, that is because let us draw the density functions of x and y.

So, this is let us say - 1, 0, 1, the density function of x looks like this. So, value is half, but y is equal to the absolute value of x. So, y cannot take negative values, because it is a function of x, what effectively happens is, whenever x takes negative value effectively, it transfers here. So, the density of y in the range of 0 to 1 is going to be > the density of x in the range of 0 to 1, in fact it is going to be exactly twice as that.

So, you will have sorry, not being exact here, the value here is going to be 1. So, that is the reason, why you have the density of y is simply equal to 1, in the range of 0 to 1 and 0 otherwise. And you can see, that this also is a valid density function and it integrates to 1.

And this is going to be a pretty powerful tool for us and there might be, there are some slightly easier tricks, that we will see later on in the course. But the functions of, the main thing, that we should take away here is that functions of random variables are also random variables.